

Jane Street Quant Trader Internship — Prep Plan

A structured 8-10 week plan. Compress it to 4 weeks if your timeline is short — just double up on the weekly blocks.

1. Understand what's actually being tested

Jane Street's process is unusual: it rewards **clear, out-loud reasoning** over polished final answers, and doesn't require any finance background. Roughly, expect this funnel:

1. **Application review** (resume + optional cover letter, reviewed by a person)
2. **Phone/video interview(s)** — probability puzzles and brainteasers, solved out loud, pen and paper only (no calculator)
3. **Superday** — a full day of back-to-back interviews: more probability/logic, and interactive **trading/market-making games** played with a trader across the table, where your score in one game often carries into the next
4. **HR/behavioral conversation** — fit, motivation, communication

Two official resources are worth going through before anything else:

- **Probability and Markets Guide** (Jane Street's own primer on the concepts they test): <https://www.janestreet.com/probability-markets/>
 - **Jane Street's mock trading interview video** and general interviewing info — start from their careers/interviewing hub: <https://www.janestreet.com/join-jane-street/>
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2. The five skill areas to train

A. Mental math (speed + accuracy under pressure)

This is a hard gate for many candidates — some rounds specifically time you on arithmetic (multiplication, division, percentages, fractions) without a calculator.

- **Zetamac** (free, the standard tool people use): <https://arithmetic.zetamac.com> — aim to comfortably clear 40-60+ correct in 2 minutes before you interview.
- Daily 10-15 min drills: 2-digit $\times$ 2-digit multiplication, percentage-of, quick fraction-to-decimal conversions.

B. Probability & expected value

The core of every round.

- **Jane Street's Probability and Markets Guide** (above) — read it twice, work every practice problem.
- Books:
 - *Fifty Challenging Problems in Probability with Solutions* — Frederick Mosteller (short, classic, exactly the puzzle style used)
 - *Heard on The Street: Quantitative Questions from Wall Street Job Interviews* — Timothy Falcon Crack (widely used across quant interviews, more finance-

flavored)

- *A First Course in Probability* — Sheldon Ross (for filling theory gaps: conditional probability, Bayes, distributions, order statistics)
- Practice the classics until you can do them fast and explain them clearly: Monty Hall, expected value of dice/coin games, urn problems, gambler's ruin, random walks.

C. Trading / market-making intuition

This is the differentiator most candidates under-prepare for.

- Practice framing a two-sided market: given a noisy signal about a value, what do you bid and what do you ask, and how wide do you need to be given your uncertainty?
- Kelly criterion and bet-sizing given an edge — know the intuition, not just the formula.
- Play informal trading/estimation games with a friend: one person picks a hidden number/range, the other has to make markets on it and adjust as new info arrives.
- **Jane Street Puzzles archive** (monthly public puzzle competition, similar flavor to interview problems): <https://www.janestreet.com/puzzles/>

D. Programming

Not required to know a specific language (Python is a safe, natural choice here since it's explicitly used in the algorithmic-trading elective).

- Be comfortable writing short, clean scripts from scratch without autocomplete — practice on a plain text editor occasionally to simulate interview conditions.
- Basic **pandas/numpy** familiarity will help you hit the ground running during the internship itself, not so much for the interview.

E. Communication under pressure

Explicitly called out in the posting ("clear and effective verbal and written communicator," "eager to ask questions, admit mistakes").

- Practice narrating your thought process out loud while solving probability puzzles — record yourself, listen back, cut the silences.
- Do this with a friend/mentor in a mock-interview format at least weekly once you're a few weeks in.
- Prepare 2–3 concrete stories (from coursework, projects, teams, jobs) that show curiosity, collaboration, and how you handled being wrong — you'll want these for the HR/behavioral conversation.

3. Week-by-week plan (8-10 weeks)

Weeks 1-2 — Foundations

- Read the Probability and Markets Guide start to finish; work every example.
- Start daily Zetamac (10 min/day).

- Begin *Fifty Challenging Problems in Probability* — 2-3 problems every other day, written out fully.

Weeks 3-4 — Build volume

- Keep Zetamac daily; track your score trend.
- Move into *Heard on The Street*; do a probability puzzle every day, timed at ~5-10 minutes.
- Start narrating solutions out loud, even alone.
- Read up on market-making basics (bid-ask spread, adverse selection, why market makers manage inventory) — a short primer is enough, you don't need textbook depth.

Weeks 5-6 — Trading intuition + mocks begin

- Introduce informal trading games with a friend (hidden number, adjust quotes as info arrives).
- Weekly mock interview with a friend/mentor: they give you a probability puzzle, you solve it out loud, they push back on your reasoning.
- Continue mental math drills; you should be near your target score by now.
- Skim Jane Street's monthly puzzles archive for extra variety.

Weeks 7-8 — Full-loop simulation

- Run a simulated Superday: back-to-back 30-45 min sessions covering (1) probability puzzle, (2) trading/market-making game, (3) behavioral questions — with a friend rotating roles.
- Tighten your resume talking points; make sure you can explain any technical/quant project on your resume in plain language.
- Draft and refine your 2-3 behavioral stories.

Weeks 9-10 (if available) — Polish

- Focus mock sessions on your weakest area from earlier feedback.
- Rest properly the 2-3 days before the actual interview — cramming late tends to hurt clarity of thought more than it helps.

4. Logistics specific to this role

- Location: Hong Kong, internship runs May-August; Jane Street covers flights and accommodation for the full program.
- You'll pick one elective (Machine Learning/Modeling, Algorithmic Trading/Market Microstructure, or Trading Strategy and Scenarios) — it's worth having a light sense of which appeals to you going in, since it shapes part of your summer, but this isn't something you're evaluated on during interviews.
- No specific degree or programming language is required — don't over-index prep on credentialing and under-index on the reasoning/communication practice above.

5. Quick reference — all links

Resource	Link
Jane Street Probability and Markets Guide	https://www.janestreet.com/probability-markets/
Jane Street Puzzles archive	https://www.janestreet.com/puzzles/
Jane Street careers / interviewing hub	https://www.janestreet.com/join-jane-street/
Zetamac mental math drill	https://arithmetic.zetamac.com
<i>Fifty Challenging Problems in Probability</i> — Mosteller	widely available in print/PDF via libraries or booksellers
<i>Heard on The Street</i> — Crack	widely available in print via booksellers
<i>A First Course in Probability</i> — Ross	standard textbook, available via libraries/booksellers
